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### 1.1.1. Calculate Momentum

01:31

Write a program that accepts the mass of an object (in kilograms) and its velocity (in meters per second), then calculates and displays the momentum of the object. The momentum is calculated using the formula:

where:

is the mass of the object (in kilograms).

is the velocity of the object (in meters per second).

#### **Input Format:**

- A single floating-point number representing the mass of the object in kilograms.
- A single floating-point number representing the velocity of the object in meters per second.

#### **Output Format:**

- The output will display calculated momentum with appropriate units (kgm/s) (rounded up to 2 decimal places).

```
mass = float(input())
velocity = float(input())
momentum = mass * velocity
print(f'{momentum:.2f}kgm/s')
```

calculate...

Submit

Explorer

Debugger

```
1 mass = float(input())
2 velocity = float(input())
3 momentum = mass * velocity
4 print(f'{momentum:.2f}kgm/s')
```

Average time  
**0.010 s**  
9.75 ms

Maximum time  
**0.012 s**  
12.00 ms

✓ 2 out of 2 shown test case(s) passed

✓ 2 out of 2 hidden test case(s) passed

✓ Test case 1 8 ms Debug

| Expected output | Actual output |
|-----------------|---------------|
| 5.0             | 5.0           |
| 10.0            | 10.0          |
| 50.00kgm/s      | 50.00kgm/s    |

✓ Test case 2 9 ms Debug

| Expected output | Actual output |
|-----------------|---------------|
| 2.3             | 2.3           |
| 4.8             | 4.8           |
| 11.04kgm/s      | 11.04kgm/s    |

Terminal

Test cases